The University of Connecticut
Department of Mechanical Engineering

ME 5130 Advanced Heat and Mass Transfer

Fall 2026 Syllabus

Instructor: Professor Wilson K. S. Chiu
Office: United Technologies Engineering Building, Room 360
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Office Hours: Before/after class, and otherwise contact the instructor to schedule.

Lecture Times:
Mondays at 5:00 – 6:15 pm ET, on-line only.
Wednesdays at 5:00 – 6:15 pm ET, in person at PWE-476 and on-line.
MS Teams Meeting Link:
https://teams.microsoft.com/meet/266146895285728?p=rCVdNojSFu2gPOlMu1
Meeting ID: 266 146 895 285 728; Passcode: BD9EX3Bx

Course Co-Requisite: *ME5507 Engineering Analysis I* (or equivalent)

Textbooks:
1. *Heat Conduction (2nd edition)*, M. N. Özişik, Wiley, 1993.
2. *Transport Phenomena (1st edition)* R. B. Bird, W. E. Stewart, E. N. Lightfoot, Wiley, (1960; *2nd edition,* 2002 and 2006 also available).

Additional References:
1. *Advanced Engineering Mathematics*, E. Kreyszig, Wiley, 2011.
2. *Fundamentals of Heat and Mass Transfer*, T. L. Bergman, and A. S. Levine, F. P. Incropera, D. P. DeWitt, Wiley, 2017.

Course Website at HuskyCT: https://learn.uconn.edu

Grading:

Homework	15%
Midterm Assessment:	30%
Final Assessment Presentation:	25%
Final Assessment Report:	30%

Midterm Assessment will cover material discussed in class, reading and homework assignments.
Final Assessment will consist of a presentation and final report.
Make-ups will not be given except in the case of *documented* medical emergencies.

Academic Conduct: All students are expected to uphold honest and ethical academic behavior. Misrepresenting mastery in an academic area (e.g., cheating), failing to properly credit information, research, or ideas to their rightful originators or representing such information, research, or ideas as their own (e.g., plagiarism) may result in disciplinary action. Link to the Academic, Scholarly, and Professional Integrity and Misconduct Policy

Authentication: UConn is required to verify that a student who registers in a course is the same student who participates in and completes the course and receives academic credit. This course will use HuskyCT as the primary student authentication method. Student authentication may also be supported by email, phone, and/or video contact. For additional information, please see https://kb.ecampus.uconn.edu/2020/12/02/authentication-of-students.

"Good-Faith" Effort: Good-faith effort is required on all assignments, projects and exams until the end of the semester. This is to help make sure that all of the learning objectives for the class are accomplished to prepare the student for future coursework. Please reach out to the instructor if you need assistance for any reason, including extensions or missing assignments. To receive a passing grade for this course, the student must complete the midterm and final assessments.

Resources for Students Experiencing Distress: The University of Connecticut is committed to supporting students in their mental health, their psychological and social well-being, and their connection to their academic experience and overall wellness. The university believes that academic, personal, and professional development can flourish only when each member of our community is assured equitable access to mental health services. The university aims to make access to mental health attainable while fostering a community reflecting equity and diversity and understands that good mental health may lead to personal and professional growth, greater self-awareness, increased social engagement, enhanced academic success, and campus and community involvement. Students who feel they may benefit from speaking with a mental health professional can find support and resources through the Student Health and Wellness-Mental Health (SHaW-MH) office at https://studenthealth.uconn.edu/mental-health. Through SHaW-MH, students can make an appointment with a mental health professional and engage in confidential conversations or seek recommendations or referrals for any mental health or psychological concern. Mental health services are included as part of the university's student health insurance plan and also partially funded through university fees. If you do not have UConn's student health insurance plan, most major insurance plans are also accepted. Students can visit the Student Health and Wellness-Mental Health located in Storrs on the main campus in the Arjona Building, 4th Floor, or contact the office at (860) 486-4705, or https://studenthealth.uconn.edu for services or questions.

Accommodations for Illness or Extended Absences: Please stay home if you are feeling ill and please go home if you are in class and start to feel ill. If illness prevents you from attending class, it is your responsibility to notify the instructor as soon as possible. You do not need to disclose the nature of your illness, however, you will need to work with the instructor to determine how you will complete coursework during your absence. If life circumstances are affecting your ability to focus on courses and your UConn experience, students can email the Dean of Students at dos@uconn.edu to request support. Regional campus students should e-mail the Student Services staff at their home campus to request support and faculty notification. COVID-19 Specific Information: People with COVID-19 have had a wide range of symptoms reported – ranging from mild symptoms to severe illness. These symptoms may appear 2-14 days after exposure to the virus and can include:

- Fever
- Cough
- Chills
- Repeated shaking with chills
- Shortness of breath or difficulty breathing
- Muscle Pain
- Headache
- Sore throat
- New loss of taste or smell

Additional information including what to do if you test positive or you are informed through contract tracing that you were in contact with someone who tested positive, and answers to other important questions can be found here: https://studenthealth.uconn.edu/updates-events/coronavirus.

TENTATIVE SCHEDULE

Week	Topic	Reading Assignment
1	Heat Conduction	Özisik Chapter 1
2	1-D Steady State Heat Conduction	Özisik Chapter 2
3	2-D Heat Conduction, Internal Heat Generation	Özisik Chapter 3
4	Mass Diffusion	BSL(1960) Chpt. 16
5	Shell Mass Balance	BSL(1960) Chpt. 17
6	Heterogeneous and Homogeneous Chemical Reactions	
7	**Midterm Assessment (October 19 or 21, 2026)**	
8	Conservation Equations	BSL(1960) Chpt. 3.1-4
9	Momentum Boundary Layer	BSL(1960) Chpt. 4.4
10	Thermal Boundary Layer	BSL(1960) Chpt. 11.4
11	Combined Heat and Mass Transfer Boundary Layers	BSL(1960) Chpt. 19.3
12	Combined HMT Boundary Layers – cont'd	
13	Combined HMT Boundary Layers – cont'd	
14	**Final Assessment Presentations and Report (December 7 or 9, 2026)**	

Revisions:
1. August 28, 2026: Version 1.
2. August 31, 2026: Updated BSL edition and years.